MATH 152 **QUIZ 3** Name Solutions

Mrs. Bonny Tighe 25 points

6.5, 7.1 TA_____ Wed 9/212022

3pts

1. Find the average value of $f(x) = \dfrac{\cos x}{1+\sin^2 x}$ *on* $[0, \pi/2]$

$$\frac{1}{\pi/2 - 0}\int_0^{\pi/2} \frac{\cos x}{1+\sin^2 x}\,dx \qquad u = \sin x \quad du = \cos x\,dx$$

$$\frac{2}{\pi}\int_0^{\pi/2} \frac{du}{1+u^2} = \frac{2}{\pi}\tan^{-1} u\Big]_0^{\pi/2} = \frac{2}{\pi}\tan^{-1}(\sin x)\Big]_0^{\pi/2}$$

$$\frac{2}{\pi}\left[\tan^{-1}(\sin \pi/2) - \tan^{-1}(\sin 0)\right] =$$

$$\frac{2}{\pi}\left[\tan^{-1}(1) - 0\right] = \frac{2}{\pi}\cdot\frac{\pi}{4} = 1/2$$

3pts

2. The linear density in a rod 8 m long is $\dfrac{10}{\sqrt{x+1}}\ kg/m$ where x is measured in meters from one end of the rod. Find the average density of the rod.

$$\frac{1}{8}\int_0^8 10(x+1)^{-1/2}\,dx = \frac{10}{8}\left[\frac{1}{1/2}(x+1)^{1/2}\right]_0^8$$

$$\frac{5}{4}\left[2\sqrt{x+1}\right]_0^8 = \frac{5}{4}\left[2\sqrt{9} - 2\sqrt{1}\right] =$$

$$\frac{5}{4}\left[6-2\right] = \frac{5}{4}(4) = 5\,kg$$

units

$\int u\,dv = uv - \int v\,du$

5pts

4. Evaluate the integrals. a) $\int t^2\ 5^{2t}\ dt$

$u = t^2 \qquad dv = 5^{2t}\,dt$

$du = 2t\,dt \qquad v = \frac{1}{2}\frac{5^{2t}}{\ln 5}$

$= t^2\left(\frac{5^{2t}}{2\ln 5}\right) - \int \frac{5^{2t}}{2\ln 5}(2t\,dt) =$

$u = t \qquad dv = 5^{2t}\,dt$

$du = dt \qquad v = \frac{5^{2t}}{2\ln 5}$

$= t^2\left(\frac{5^{2t}}{2\ln 5}\right) - \frac{1}{\ln 5}\int t\,5^{2t}\,dt$

$= \frac{t^2(5^{2t})}{2\ln 5} - \frac{1}{\ln 5}\left[t\left(\frac{5^{2t}}{2\ln 5}\right) - \int \frac{5^{2t}}{2\ln 5}\,dt\right]$

$= \frac{t^2 5^{2t}}{2\ln 5} - \frac{t\,5^{2t}}{2(\ln 5)^2} + \frac{1}{2(\ln 5)^2}\int 5^{2t}\,dt$

$= \frac{t^2 5^{2t}}{2\ln 5} - \frac{t\,5^{2t}}{2(\ln 5)^2} + \frac{1}{2(\ln 5)^2}\cdot\frac{5^{2t}}{2\ln 5} + C$

b) $\int \cos(\ln x)\,dx$

$u = \cos(\ln x) \qquad dv = dx$

$du = -\sin(\ln x)\frac{1}{x}\,dx \qquad v = x$

$= x\cos(\ln x) - \int x\left(-\sin(\ln x)\frac{1}{x}\right)dx$

4pts

$x\cos(\ln x) + \int \sin(\ln x)\,dx$

$u = \sin(\ln x) \qquad dv = dx$

$du = \cos(\ln x)\frac{1}{x}\,dx \qquad v = x$

$x\cos(\ln x) + x\sin(\ln x) - \int x\cos(\ln x)\frac{1}{x}\,dx$

$\int \cos(\ln x)\,dx = x\cos(\ln x) + x\sin(\ln x) - \int \cos(\ln x)\,dx$

$+1 \qquad +$ ☺

$2\int \cos(\ln x)\,dx = x\cos(\ln x) + x\sin(\ln x)$

$\int \cos(\ln x)\,dx = \boxed{\frac{1}{2}\left(x\cos(\ln x) + x\sin(\ln x)\right) + C}$

$\int t^2 5^{2t}\,dt = \frac{t^2 5^{2t}}{2\ln 5} - \frac{t\,5^{2t}}{2(\ln 5)^2} + \frac{5^{2t}}{4(\ln 5)^3} + C$

Intermediate substitution

4 pts c) $\int \sin\sqrt{x}\,dx$

$m = \sqrt{x}$ $\quad dm = \frac{1}{2\sqrt{x}}dx$

$2\sqrt{x}\,dm = dx \quad = 2m\,dm$

$\int \sin m(2m\,dm)$

$u = 2m \quad dv = \sin m\,dm$

$du = 2\,dm \quad v = -\cos m$

$-\cos m(2m) - \int -\cos m(2\,dm)$

$-2m\cos m + 2\int \cos m\,dm =$

$-2m\cos m + 2\sin m + C$

$-2\sqrt{x}\cos\sqrt{x} + 2\sin\sqrt{x} + C$

4 pts d) $\int \sin^{-1} x\,dx$

$u = \sin^{-1} x \quad dv = dx$

$du = \frac{1}{\sqrt{1-x^2}}dx \quad v = x$

$= x\sin^{-1}x - \int x\left(\frac{1}{\sqrt{1-x^2}}dx\right)$

$u = 1-x^2$

$du = -2x\,dx$

$\frac{du}{-2x} = dx$

$= x\sin^{-1}x - \int \cancel{x}(u^{-1/2})\left(\frac{du}{-2\cancel{x}}\right)$

$= x\sin^{-1}x + \frac{1}{2}\int u^{-1/2}\,du$

$= x\sin^{-1}x + \frac{1}{2}\cdot\frac{1}{1/2}u^{1/2} + C$

$= x\sin^{-1}x + \sqrt{1-x^2} + C$

MATH 152 **QUIZ 3A** Name Solution

Mrs. Bonny Tighe 25 points

6.5, 7.1 TA_____ Wed 9/21/2022

3pts

1. Find the average value of $f(x)=\dfrac{2^x}{1+2^{2x}}$ *on* [0,1]

$m = 2^x$

$dm = 2^x \ln 2\, dx$

$= dx$

$\dfrac{dm}{m \ln 2} = \dfrac{dm}{2^x \ln 2}$

$\dfrac{1}{1-0}\int_0^1 \dfrac{2^x}{1+2^{2x}}\,dx$

$\int \dfrac{m}{1+m^2}\left(\dfrac{dm}{m \ln 2}\right) = \int \dfrac{1}{\ln 2}\,\dfrac{dm}{1+m^2} = \dfrac{1}{\ln 2}\tan^{-1} m + C$

$\dfrac{1}{\ln 2}\tan^{-1}(2^x) + C$

2. A cup of coffee has temperature 95° C and takes 30 minutes to cool to 61° C in a room with temperature 20° C. By Newton's Law of Cooling, the temperature of the coffee after t minutes is $T(t) = 20 + 75e^{-kt}$ where k = 0.05. What is the average temperature of the coffee during the first half hour?

3pts

$\dfrac{1}{30-0}\int_0^3 \left(20 + 75e^{-0.05t}\right) dt$

$.05 = \dfrac{5}{100} = 1/20$

$.05(30) = 1.50$

$\dfrac{1}{30}\left[2t + \dfrac{75e^{-0.05t}}{-0.05}\right]_0^{30}$

$\dfrac{1}{30}\left[2t + 75e^{-0.05t}(20)\right]_0^{30}$

$\dfrac{1}{3}\left[60 + 75(20)e^{-0.05(30)} - 75e^0(20)\right]$

$\dfrac{1}{3}\left[60 + 1500e^{1.5} - 75(20)\right]$ degrees/minute

$\dfrac{1}{3}\left[60 + 1500e^{1.5} - 1500\right]$ deg/min

4. Evaluate the integrals. a) $\int t^2 e^{-2t}\, dt$

$u = t^2 \quad dv = e^{-2t}\,dt$
$du = 2t\,dt \quad v = -\frac{1}{2}e^{-2t}$

4 pts

$= -\frac{1}{2}e^{-2t}t^2 - \int -\frac{1}{2}e^{-2t}\, 2t\,dt$

$u = 2t \quad dv = e^{-2t}$
$du = 2\,dt \quad v = -\frac{1}{2}e^{-2t}$

$= -\frac{1}{2}t^2e^{-2t} + \frac{1}{2}\left[-\frac{1}{2}e^{-2t}(2t) - \int -\frac{1}{2}e^{-2t}\, 2\,dt\right]$

$= -\frac{1}{2}t^2e^{-2t} - \frac{1}{2}te^{-2t} + \int e^{-2t}\,dt$

$= -\frac{1}{2}t^2e^{-2t} - \frac{1}{2}te^{-2t} - \frac{1}{2}e^{-2t} + C$

5 pts b) $\int e^{2x}\sin(3x)\,dx$

$u = \sin(3x) \quad dv = e^{2x}dx$
$du = 3\cos(3x) \quad v = \frac{1}{2}e^{2x}$

$\frac{1}{2}e^{2x}\sin(3x) - \int \frac{1}{2}e^{2x}(3)\cos(3x)\,dx$

$\frac{1}{2}e^{2x}\sin(3x) - \frac{3}{2}\left[\int e^{2x}\cos(3x)\,dx\right]$

$u = \cos(3x) \quad dv = e^{2x}dx$
$du = -3\sin(3x) \quad v = \frac{1}{2}e^{2x}$

$\frac{1}{2}e^{2x}\sin(3x) - 3/2\left[\cos(3x)\frac{1}{2}e^{2x} - \int \frac{1}{2}e^{2x}(-3\sin(3x))\,dx\right]$

$\frac{1}{2}e^{2x}\sin(3x) - \frac{3}{4}e^{2x}\cos(3x) - \frac{9}{4}\int e^{2x}\sin(3x)\,dx$

$+9/4$

$+9/4$

$13/4\int e^{2x}\sin(3x)\,dx = \frac{1}{2}e^{2x}\sin(3x) - \frac{3}{4}e^{2x}\cos(3x)$

$\int e^{2x}\sin(3x)\,dx = \frac{4}{13}\left[\frac{1}{2}e^{2x}\sin(3x) - \frac{3}{4}e^{2x}\cos(3x)\right] + C$

4 pts c) $\int \cos\sqrt{x}\,dx$

intermediate sub

$m = \sqrt{x}$

$dm = \frac{1}{2\sqrt{x}}dx = \frac{1}{2m}dx$

$dm\,(2m) = dx$

$\int \cos m \; 2m\,dm$

parts

$u = 2m$ $\quad dv = \cos m\,dm$

$du = 2dm$ $\quad v = \sin m$

$2m \sin m - \int \sin m\,(2dm) = 2m\sin m + 2\cos m + C$

$2\sqrt{x}\sin\sqrt{x} + 2\cos\sqrt{m} + C$

4 pts d) $\int \ln x\,dx$

$u = \ln x$ $\quad dv = dx$

$du = \frac{1}{x}dx$ $\quad v = x$

$= x\ln x - \int dx$

$x\ln x - \int x\left(\frac{1}{x}dx\right)$

$x\ln x - x + C$